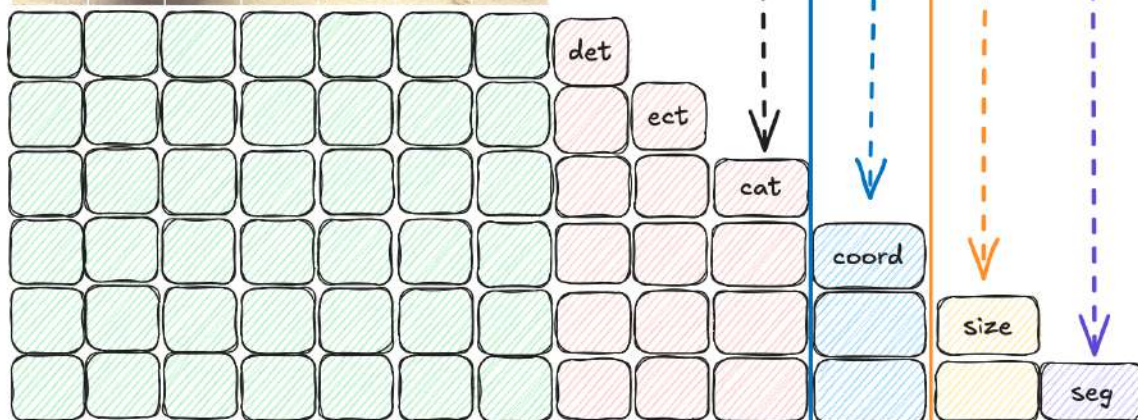
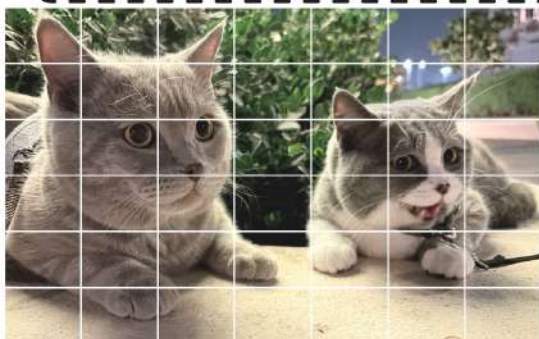


Dense, early-fusion, autoregressive Transformer

Patch embedding

Token embedding

joint attention with image block mask + causal mask



Language modeling head

Projection

Center decoder

Size decoder

Center encoder

Size encoder

coord

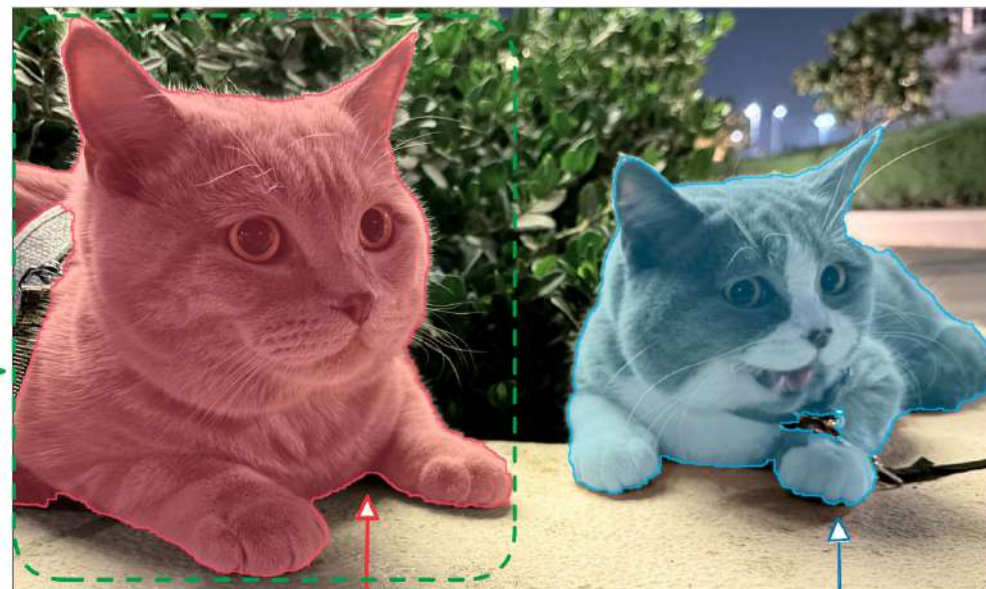
size

seg

coord

size

seg



dot

dot

Image Feature Upsampler



high resolution image feature